Jupiter's four largest moons are called Galilean moons. There are 63 others.

Europa. For the longest time scientist didn't know much about Europa - it was just a moon like any other. But with the launch of Jovian probes in 1977 and 1989 in the Voyager and Galileo missions respectively, they began to get a clearer idea of what made Europa special. The moon was determined to be the sixth largest moon in the Solar System and composed primarily of silicate rock. The atmosphere was found to be made up almost entirely of oxygen, albeit in a tenuous form.

Data from these missions also gave scientists enough evidence to posit that Europa is in fact a moon of salt water encased in a shell of ice. This hypothesis was further solidified as new evidence of its surface indicated that the outer crust was a free floating ice shelf that included clay like minerals called phyllosilicates. The discovery of phyllosilicates is significant since it has always been associated with organic materials. Given the discoveries of phyllosilicates and sub-surface water the conclusion became very simple - potential life.

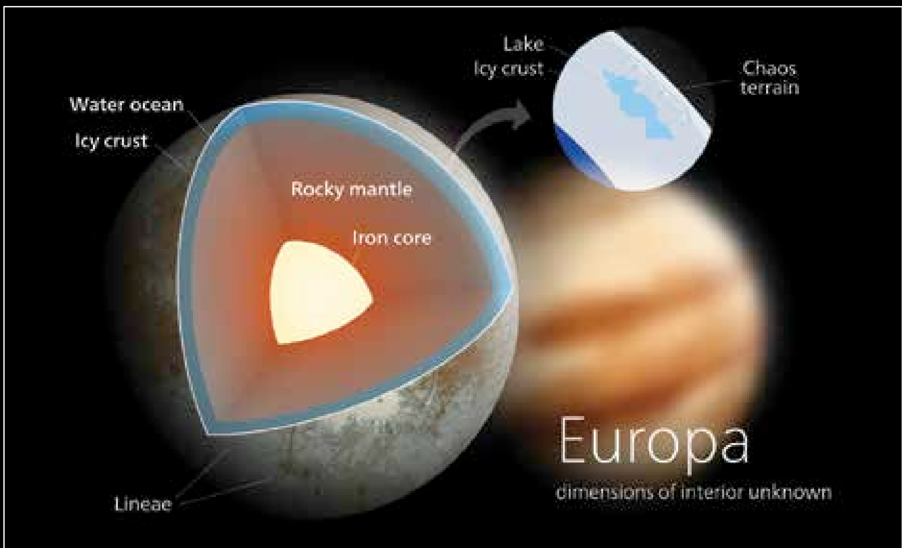
The search for life begins with the search for water

Life in the cosmos has so far been the rarest observation. Only on Earth do we find it, and it is everywhere on the planet. But what about out there in space? For decades we have sent out radio waves and space probes, hoping to stumble upon some evidence of life intelligent enough to talk back. We have even salvaged rocks and dirt from distant planets to sift for some sign of ancient microbial life gone extinct. And through it all we have found no hope. After all, life as we know it requires a unique and often rare alchemy of circumstances to arise. The origins of life requires the right mix of temperature, fluidity and organic elements. And given Europa's abundance of subsurface water and other seductive factors, we can see why Europa is so tremendously tantalising.

The current data on Europa has indicated that under the ice shell surface lies a very deep ocean. The early probes sent showed that the moon's icy covering varied in thickness anywhere from 1.5 kilometers to 29 kilometers. This icy surface was seen as being near pristine, with no impact craters, and being relatively smooth. This could only be if the moon is young and active and still experiencing sub-surface volcanic activity which keeps renewing the surface ice. This finding was further supported by the aberrant magnetic field of the moon which could only be explained by an electrically conductive fluid beneath the surface - salt water.

While the probable presence of water on the moon doesn't distinguish it from other extraterrestrial objects in the Solar System it does have other unique traits. The amount of radiation the moon receives from Jupiter is highly persistent and powerful. At 540 rads of output, the compounds on the surface of the moon are reduced to simpler essential elements such as oxygen and hydrogen. The radiation converts the water to hydrogen and oxygen which combine with surface elements to form carbon dioxide, hydrogen peroxide and other compounds essential for life.

Other geological evidence from Europa also suggests that the moon has a hot core. The presence of a heated rocky mantel provides the moon with the energy needed for organic life to emerge. The unique effects of Jupiter's powerful tidal forces on the moon during its 84 hour orbit causes the water to be in constant motion preventing freezing below the surface. The combination of these key elements - energy, water



The ice shell of Europa is theorised to act like Earth's tectonic plates moving over a salt water ocean.